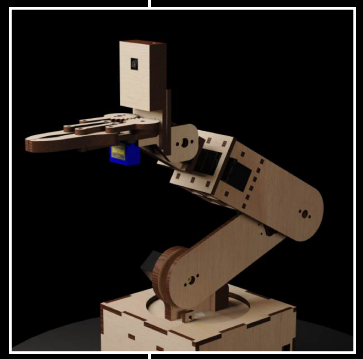
theo bui

Designer

Engineer

Computer Scientist

Contents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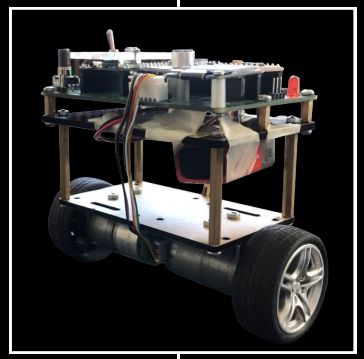
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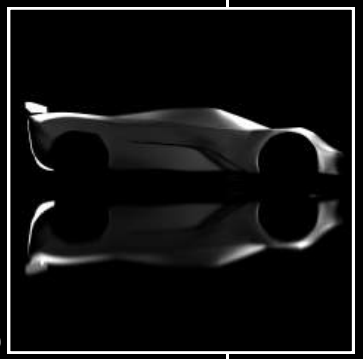
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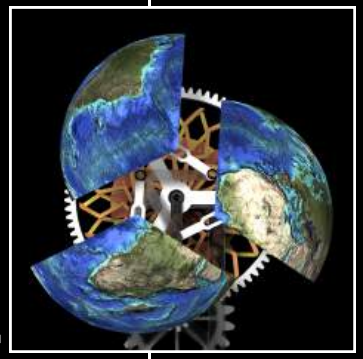
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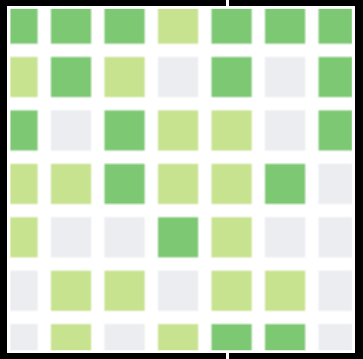
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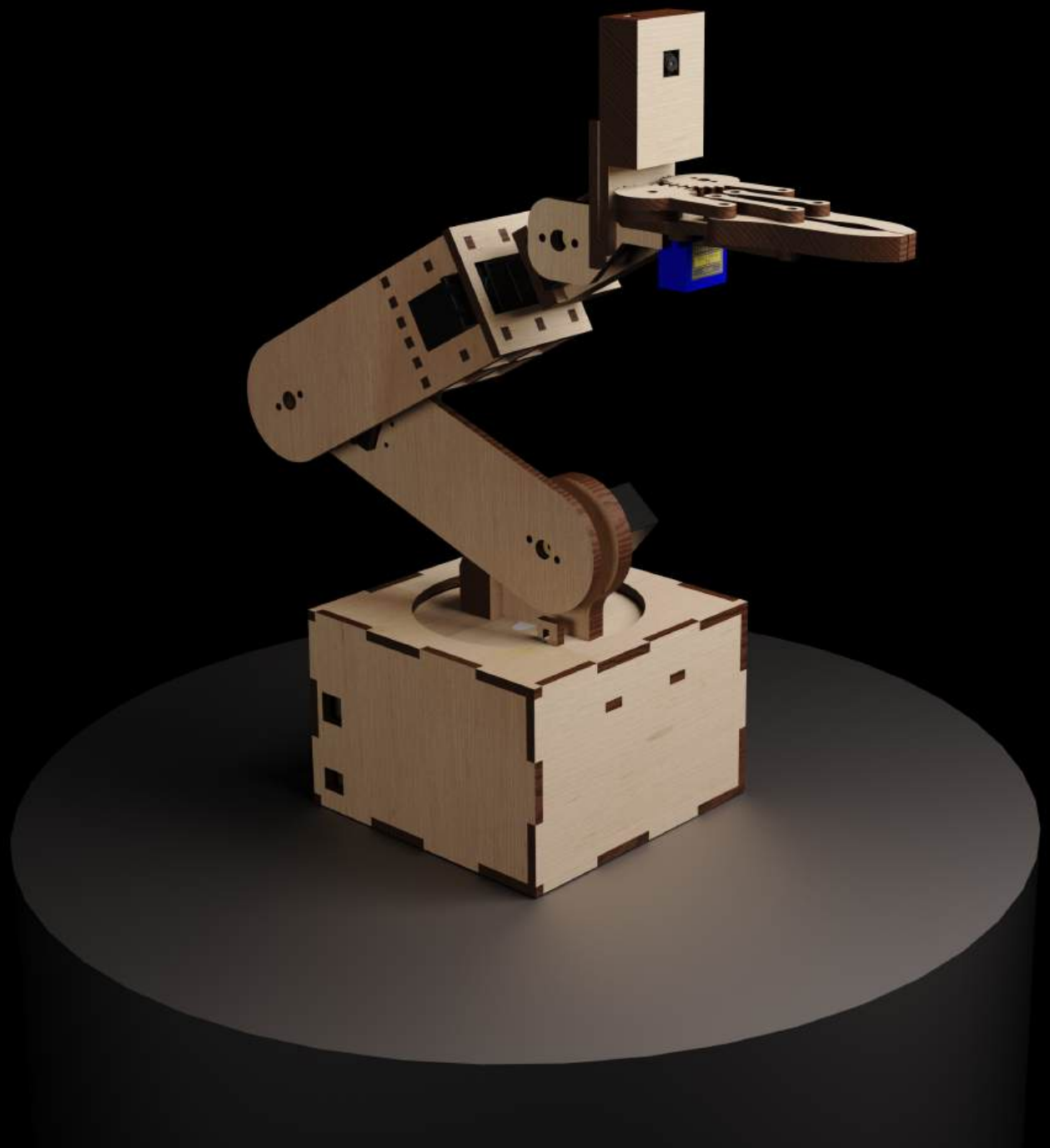
Arm-e

face tracking,
5-axis,
clamping,
remote controlled,
robot arm

This face-tracking robotic arm was the focus of a GIZMO project. We were tasked to construct an interactive robot made of 3mm plywood.



Arm-e

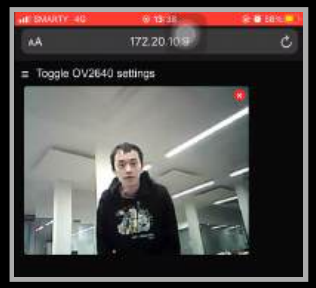
Iterative design method

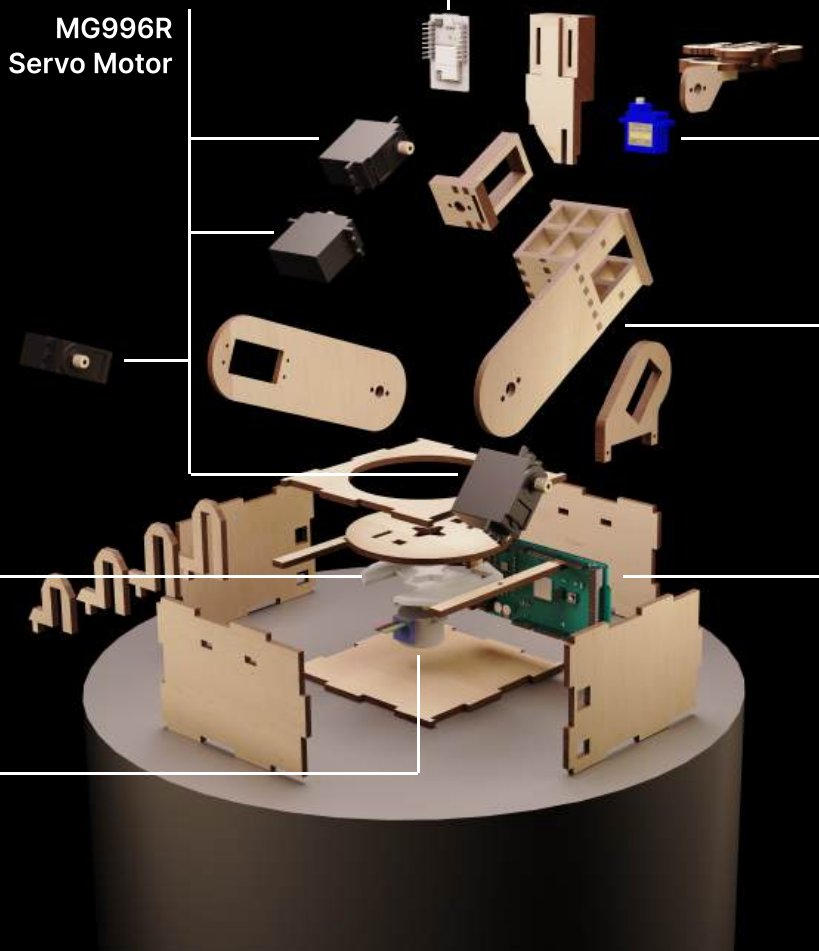
Using initial sketches, a CAD model was created. This was used to lasercut the plywood peices and assemble the arm.

Many iterations of this process followed as a result of unanticipated design flaws such as:

- *Addition of supports and counter weights preventing the arm from toppling over*
- *Addition of ports to connect directly to the boards without removing the base*
- *Redesign of links and servo supports halving the amount of plywood used and reducing the torque required by the stepper motor*

The arm's interaction included an ESP32 Camera that enabled facial detection and tracking using the various servo and stepper motors. The robot only moved to track my face, not anyone else's.

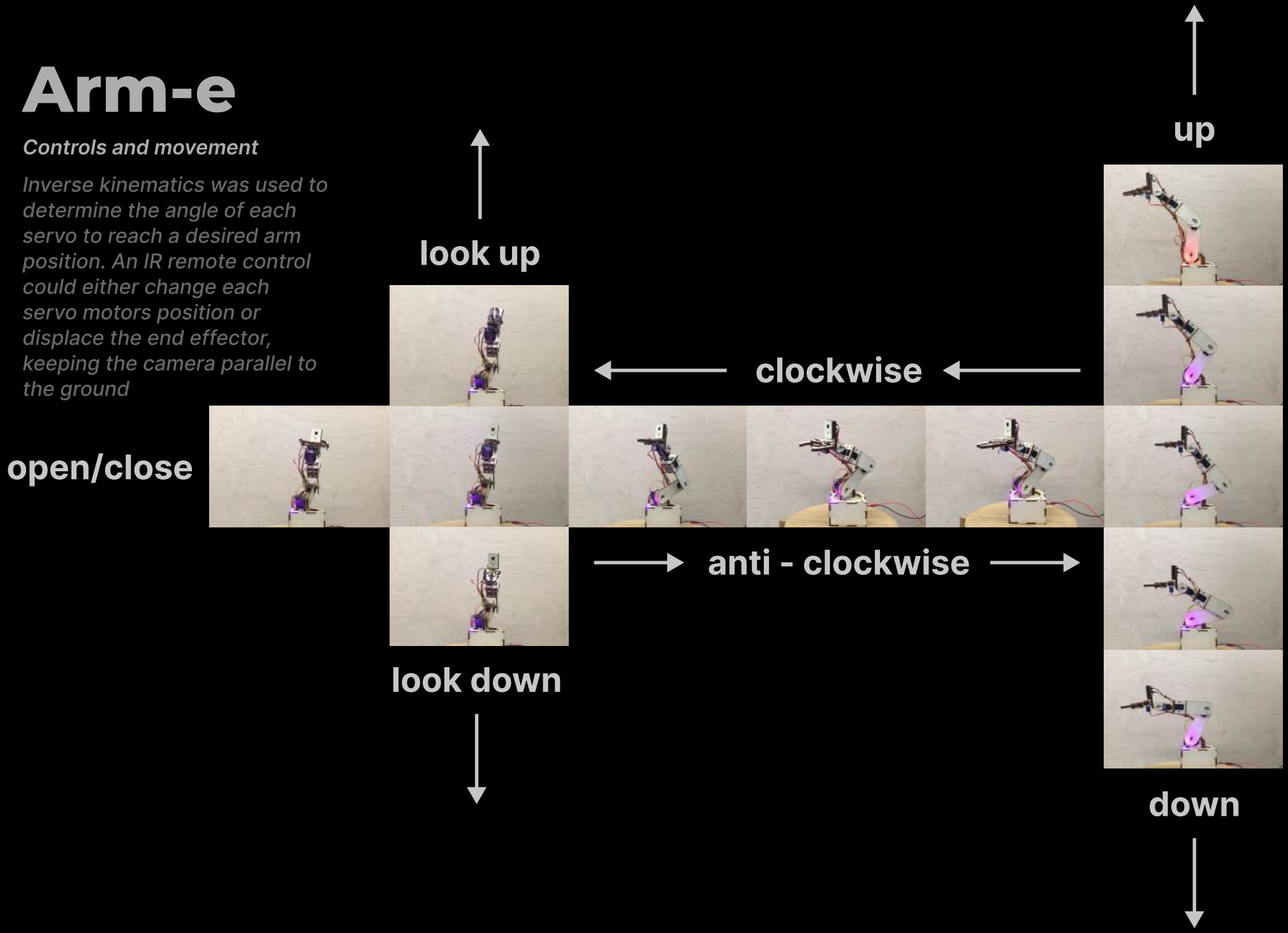




Arm-e

Controls and movement

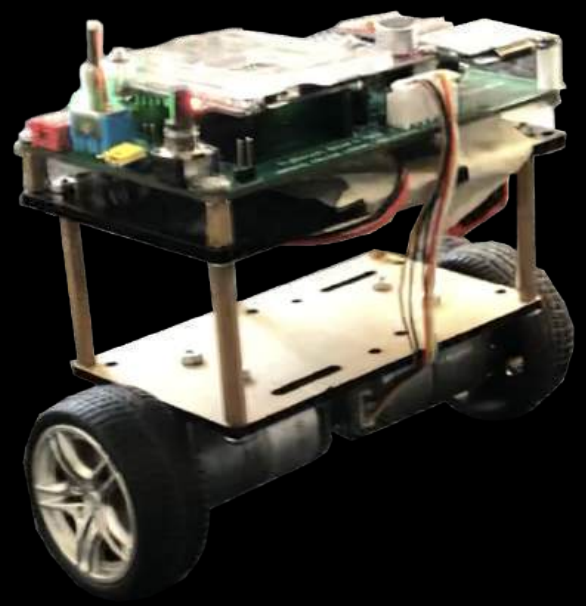
Inverse kinematics was used to determine the angle of each servo to reach a desired arm position. An IR remote control could either change each servo motors position or displace the end effector, keeping the camera parallel to the ground

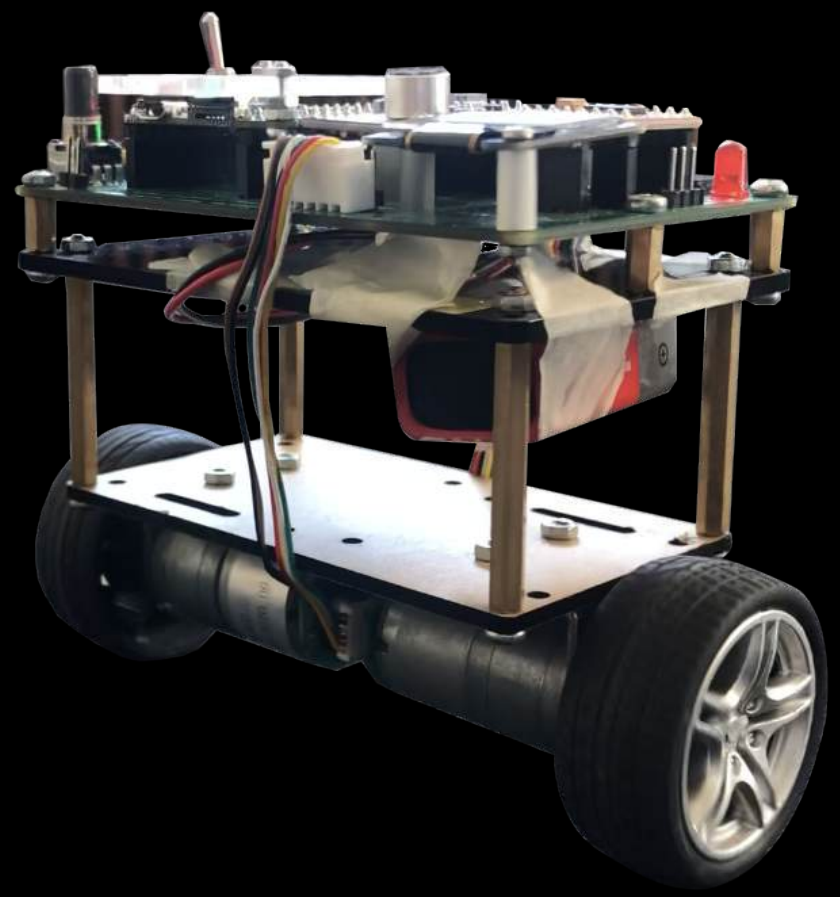


Balancing Segway

Dancing segway feedback controlled via PID

Exploring feedback control with a segway using 2 motors, a gyroscope, an accelorometer, a microphone, and a PyBench. This project tested my patience and dedication when tuning the PID coefficients which took 10+ hours.





Balancing Segway

Signal Processing

Using an integrated microphone, a beat detection algorithm allows the segway to complete predetermined moves to the beat of a song.

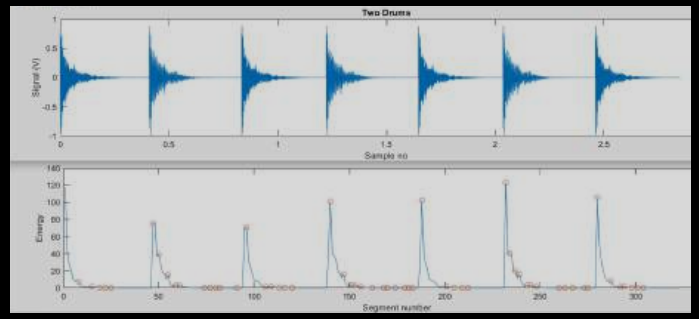
Pointers were used to add all instantaneous energies comparing each value to the average engery. If the energy ratio was sufficient, then a beat was detected and the robot preformed a “dance” move.

Feedback Control

After understanding the implications of different feedback controllers, a PID controller was chosen. To implement this, error had to be calculated exactly.

Determining the pitch, the angle of the segway from its vertical axis (the error), was done by a gyroscope and accelerometer. Due to the high frequency noise from the gyroscope and the low frequency drift from integrating the accelerometer data, a complementary filter was used.

Beat Detection



Beat Detection Code

```python
while True:             # Main program loop
    if audio.buffer_is_filled():        # semaphore signal from ISR - set if buffer is full

        # Fetch instantaneous energy
        E = audio.inst_energy()         # fetch instantenous energy
        audio.reset_buffer()            # get ready for next epoch

        # compute moving sum of last 50 energy epochs with circular buffer
        sum_energy = sum_energy - e_buf[e_ptr] + E
        e_buf[e_ptr] = E            # over-write earliest energy with most recent
        e_ptr = (e_ptr + 1) % M     # increment e_ptr with wraparound - 0 to M-1
        average_energy = sum_energy/M

        # Compute ratio of instantaneous energy/average energy
        c = E/average_energy
        if (pyb.millis()-tic > MIN_BEAT_PERIOD):    # If longer than minimum period
            if (c>BEAT_THRESHOLD):      # look for a beat

                b_LED.on()
                pyb.delay(10)
                b_LED.off()

                dd = random.randint(0, 4)

                if (dd == 0):
                    Forward(50)
                elif (dd == 1):
                    Backward(50)
                elif(dd==2):
                    Left(50)
                elif(dd==3):
                    Right(50)
                elif(dd==4):
                    Stop()

                tic = pyb.millis()      # reset tic
        buffer_full = False             # reset status flag
```

PID Control

```python
class PID(object):
    def __init__(self, kp, kd, ki):
        self.cumulative_pitch_error = 0
        self.prev_error = 0
        # self.target = target
        self.kp = kp
        self.kd = kd
        self.ki = ki

    def getPWM(self, pitch, pitch_dot, target, dt):

        if (dt == 0):
            # print("no time")
            return 0

        error = pitch - target

        if (abs(error) <= 0.2):
        # if (abs(error) <= 1.1):
            # print("upright")
            return 0

        proportional = self.kp*error

        # # kd part - differential
        differential = self.kd * pitch_dot

        self.cumulative_pitch_error += error * dt*0.001

        # ki part - integral
        integral = self.ki*self.cumulative_pitch_error

        # summation
        w = proportional+differential+integral
```

Aerodynamic Design

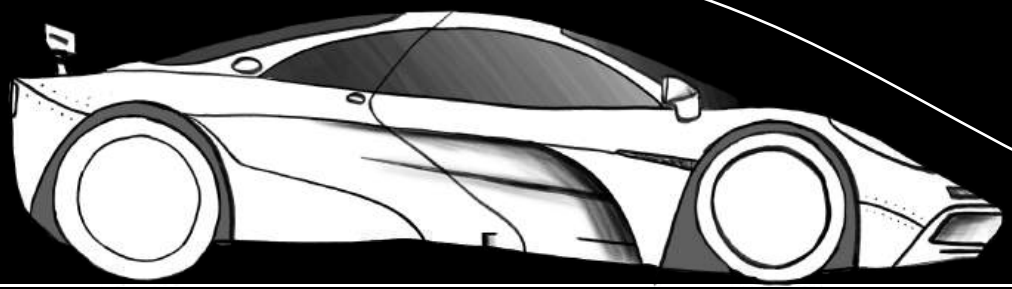
This project investigated the impact of design choices on the car's aerodynamic characteristics.

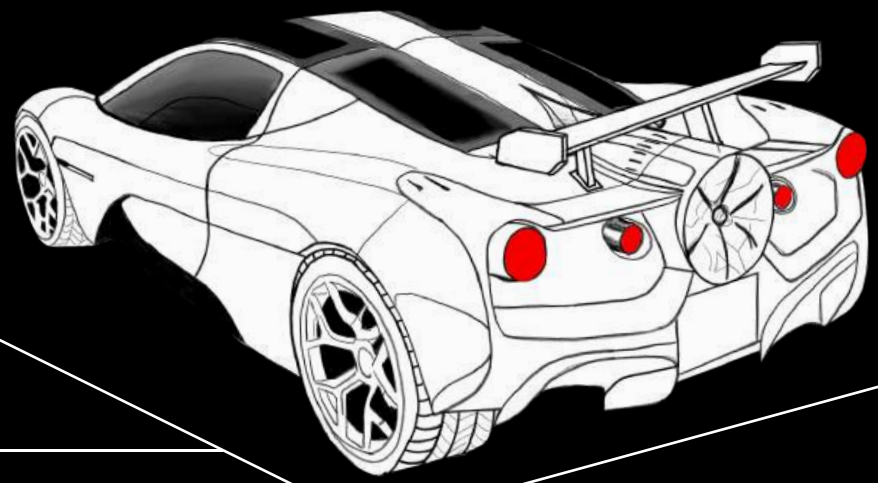
From sketches, to Solidworks and Fusion 360 CAD softwares, a car was constructed.
Solidworks' 2021 Flow Simulation was used for the anaysis.

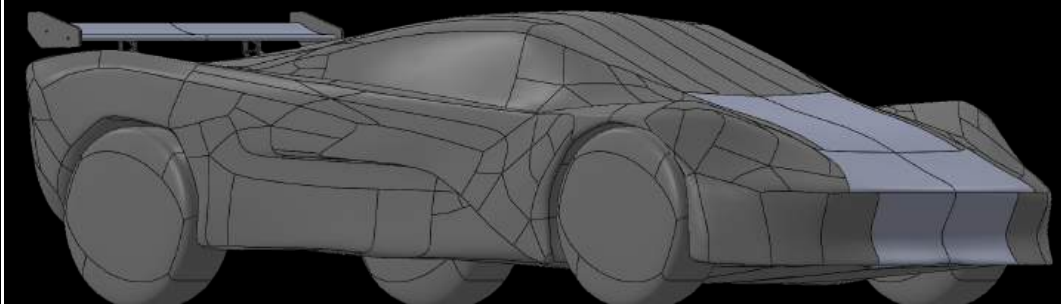
Aerodynamic Design

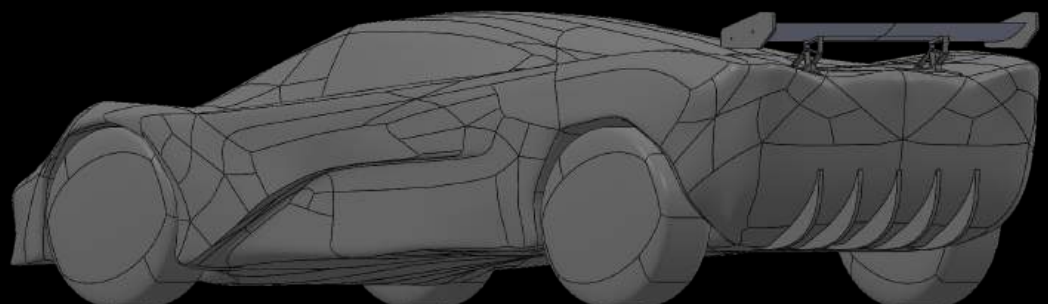
CAD model construction from sketches

Fusion 360's surface modelling feature was used to achieve smooth surfaces. The model was then exported into SolidWorks 2021 for the analysis









Aerodynamic Design

Flow and Pressure Simluations

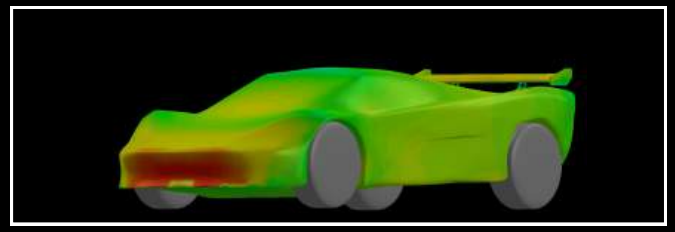
Flow trajectories allows the visualisation of the displacement of the air as the car moves. Quanitites such as velocity and direction of air particles help determined the pressure differentials that affect downforce.

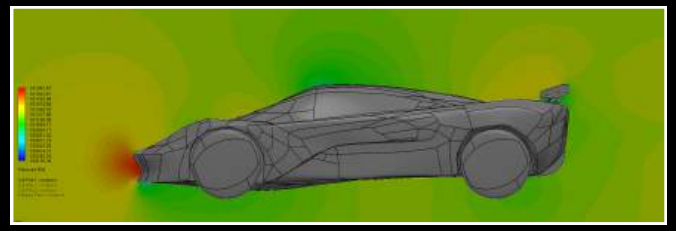
Vorticies and flow reattachment were investigated to measure turbulent flow's affect on forces acting on the car.

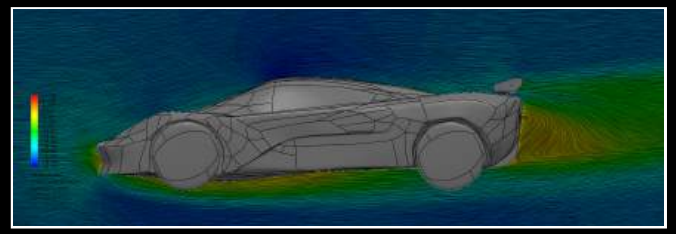
The distribution of pressure on the car's surface helps determine areas of improvement for ground effect and lift experienced on the rear of the car.

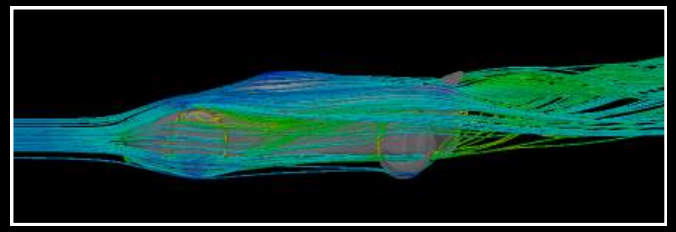
The efficacy of the venturi tunnel on the underside of the car is limited due to the run off on either side of the car. Combing this with the flow simulation analysis, vorticies off the side of the car would create "air curtains" constraining the air to experience the venturi tunnel.

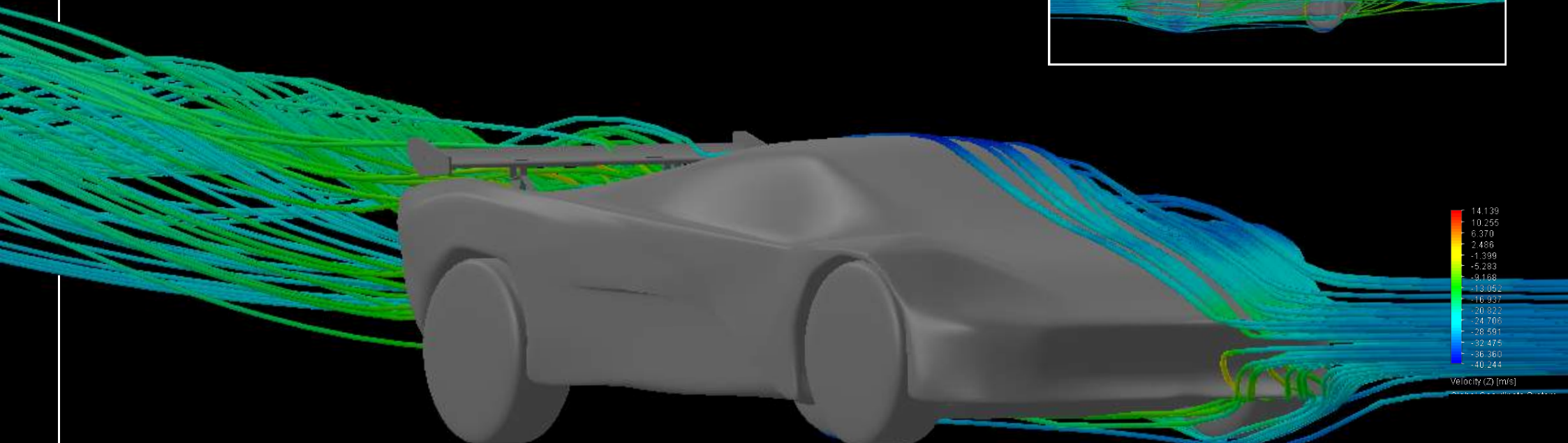
The lack of internal cavities limits the analysis creating an excessive pressure on the grill as well as the lack of air curtains and flow reattachment on the side of the wheels.











World On Fire

100% 3d printed linkage mechanism

This dynamic structure represents the cause and effect of climate change symbolically

This project demonstrates the CAD skills required for a functional mechanism. It includes various gears, snap fits, sliders, and rotational joints. The physical model is 100% 3D printed.

World On Fire

Linkage movement

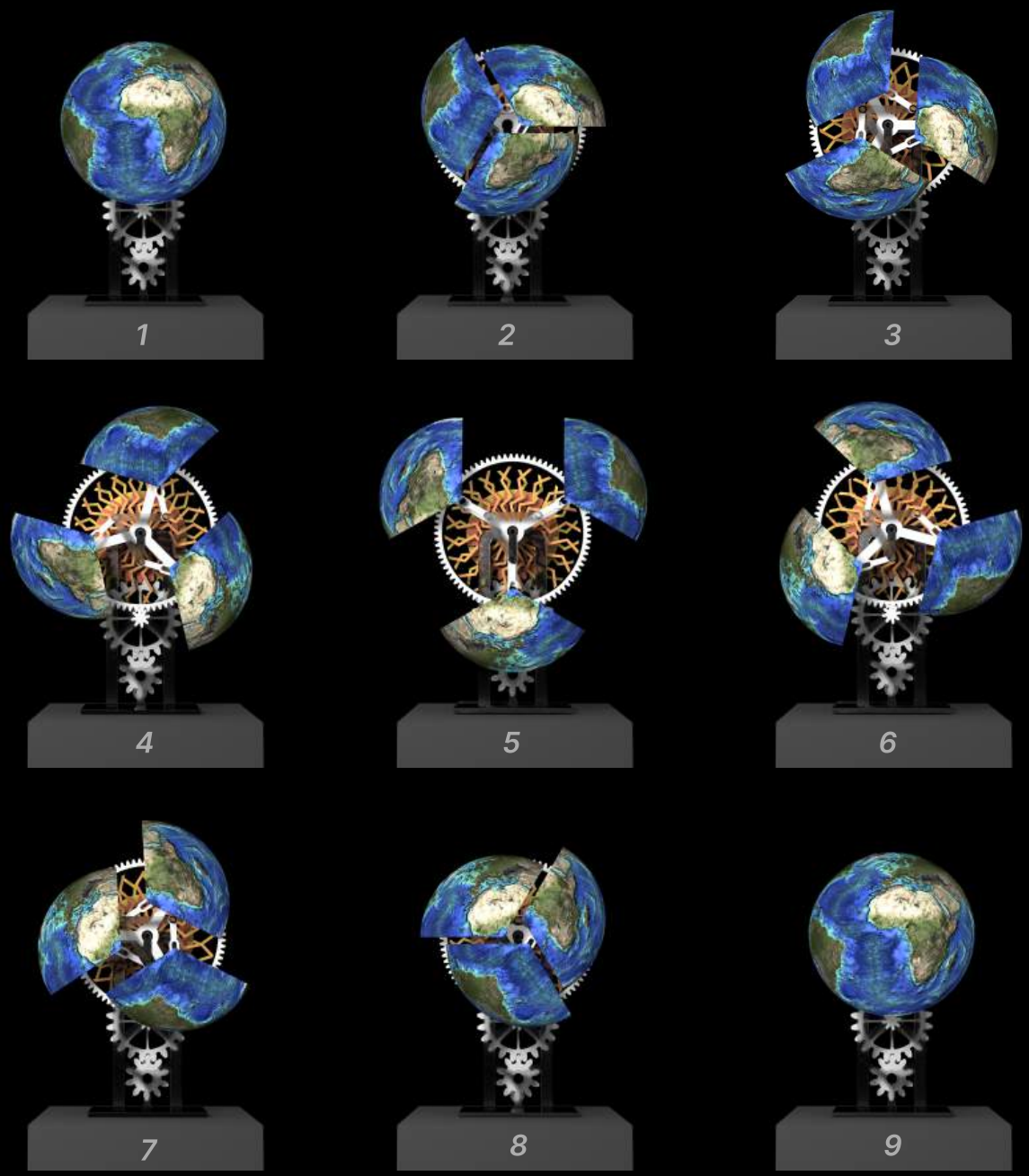
Through the continous movement of the hand crank, the world explodes. At the center of the explosion is a roaring rotating fire.

The motion is continuous and repeats endlessly.

The difficulty of using additive manufacturing (with a 3d printer) for a project is its manufacturability. Anything can be built on CAD software but not all physical pieces can fit into others or can be printed (smoothly), even with supports. The design of pieces need to consider the limitations of the printer.

Tolerances had to be tried and tested to achieve the proper support and contact, allowing rotation of the gears.

Due to the expansion of the filament during printing, snap fits were inconsistent, thus filing and the addition of material using a soldering iron and PLA filament compensated the printers inaccuracy.



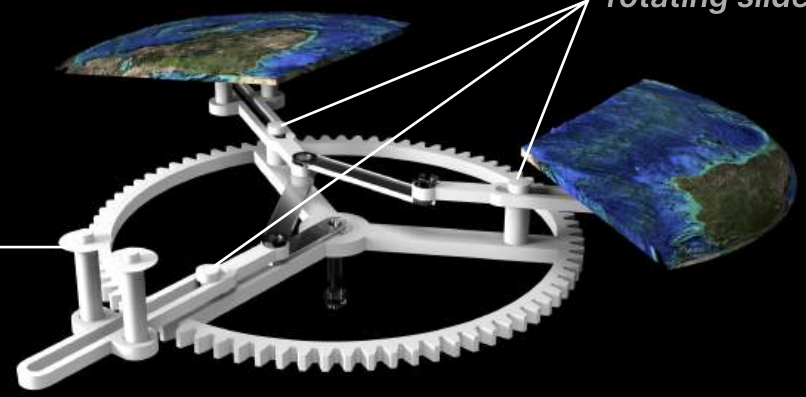
World On Fire

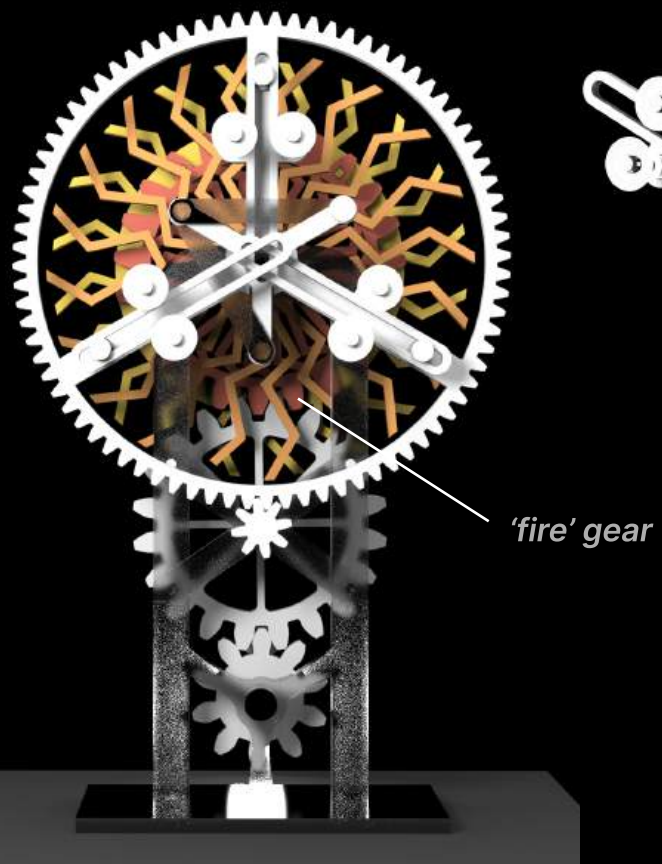
Mechanism

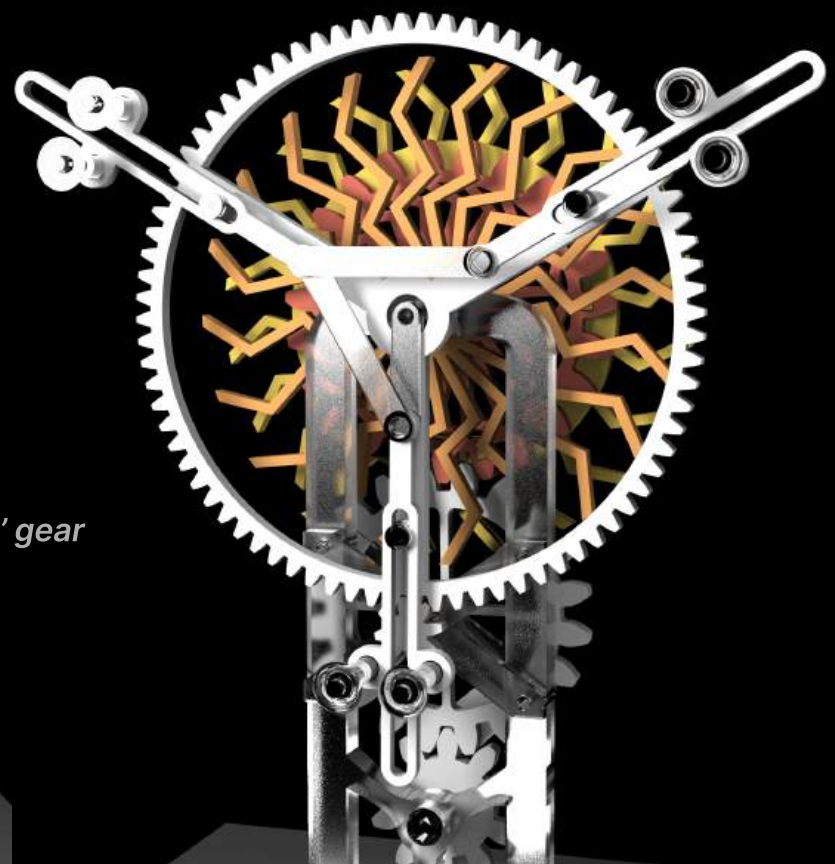
The entire mechanism is stimulated by the rotation of a hand crank. The 10-teeth crank acts upon a 20-teeth gear connected to a 8-teeth gear, finally connected to a 65-teeth gear.

As the largest gear turns, the rotating sliders displace the 3-pieced earth. All sliders (and pieces) follow the same continous movement, only rotated 120 degrees for each piece of the earth.

As this happens, the hand crank's gear, directly connected to the red gear, rotates with the yellow loop. The triangular design of the orange, red and yellow parts paired with their opposite rotations offer a hypnotizing view resembling flames.







Carbonless

A passive carbon tracker that teaches and rewards sustainable behavior

Award winning app concept:

Hero Labs: "Using Technology to make a meaningful impact" Prize

The Mayor's Entrepreneur Competition: Semi Finalist

GitHub

Repository of my computing projects

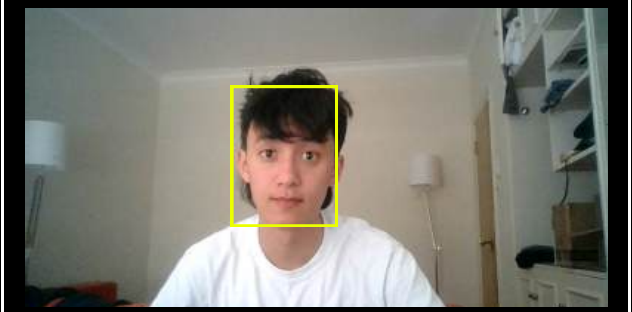
MNIST-Convolutional-Neural-Network

Chess-Game

Sudoku-Solver

Self-Balancing-Segway-PID

Face-Tracking-Robotic-Arm



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